Activity 1

Brightlearn
SQL Fundamentals

Question 1
Write a SQL query to retrieve all columns from the employees table

```
Select *
FROM employees_db. employees
LIMIT 5;
```

employee

id	First_nam	Last_name	department	Salary	hire_date	City
1	John	Doe	IT	55 000	2018-06-15	New York
2	Jane	Smith	HR	48 000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60 000	2017-09-30	Los Angels
4	Sarah	Brown	IT	53 000	2021-01-10	New York
5	David	White	Marketing	52 000	2016-04-10	San Francisco

Question 2 Select Distinct
Write a SQL query to find all the unique departments in the employees table

```
SELECT DISTINCT
        ~~Dep~~ department
FROM employees_db.employees;
```

Employees

department
IT
HR
Finance
Marketing

3 ORDER BY Statement

Write a SQL query to retrieve all employees' first and last names, ordered by salary in descending order

```
SELECT first_name,
       last_name,
       Salary
FROM employees_db.employees
ORDER BY Salary DESC;
```

Employees

first_name	last_nam	Salary
Emily	Davis	62000
Mike	Johnson	60 000
Robert	Wilson	59000
John	Doe	55 000
Sarah	Brown	53000
Daniel	Clark	53 000
David	White	52000
Jessica	Moore	51000
Laura	Hall	50000
Jane	Smith	48000

4 Limit Statement

Write a SQL query to retrieve the top 5 highest-paid employees

```
SELECT *
FROM employeedb.employees
ORDER BY Salary DESC,
Limit 5;
```

Employees

id	First_name	last_name	department	Salary	hire_date	City
	Emily	Davis	IT	62000	2015-02-14	Chicago
	Mike	Johnson	Finance	60 000	2017-09-30	Los Angeles
	Robert	Wilson	Finance	59 000	2019-10-01	Houston
	John	Doe	IT	55 000	2018-06-15	New York
	Sarah	Brown	IT	53 000	2021-03-25	New York

5 Where Statement

Write a SQL query to find employees who work in the IT department

```
Select *
FROM employee_db..employees
Where department = 'IT';
```

Employee

id	first_name	last_name	department	Salary	hire_date	City
1	John	Doe	IT	55 000	2018-06-15	New York
4	Sarah	Brown	IT	53 000	2021-03-25	New York
6	Emily	Davis	IT	62 000	2015-02-14	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

6 AND Statement

Write a SQL query to find employees who work in the finance departmend ~~and~~ AND have a salary greater than 58000

```
Select first_name,
       Last_name,
       department,
       Salary
From employee_db.employees
WHERE department = Finance AND salary < 58 000;
```

Emeployees

First_name	Last_name	department	Salary
Mike	Johnson	Finance	60 000
Robert	wilson	Finance	59 000

7 OR Statement

Write a SQL query to find employees who work in the HR department OR the marketing department

```
Select  id,
        department
FROM    employee_db.employee
WHERE   department = Finance OR department = Marketing
;
```

Employee

id	department
2	HR
8	HR
5	Marketing
9	Marketing

8 NOT Statement

Write a SQL query to find employees who do not work in the IT department

```
Select  id,
        First_name,
        department
FROM employee_db.employee
WHERE NOT department = IT department;
```

Employee

id	First_name	department
2	Jane	HR
3	Mike	Finance
5	David	Marketing
7	Robert	Fanance
8	Jessica	HR
9	Daniel	Marketing

9 IN statement

Write a SQL query to find employees who are in the HR, IT or Finance departments

```
Select department
FROM employee_db.employee
WHERE department IN('HR', 'Finance', 'IT');
```

Employee

department
IT
HR
Finance
IT
IT
Finance
HR
IT

10 Combining Condition

Write a SQL query to find employees who are in the IT department, have a salary greater than 50 000 and are located in New York

```
Select   Salary
~~Select~~ First_name
         department
         City
From     employee_db . employee
WHERE    department = 'IT' AND 'Salary' > 50 000 AND location =
         'New York';
```

First_name	department	City	Salary
John	IT	New York	55 000
Sarah	IT	New York	53 000

11 Combining WHERE, AND, and ORDER BY

Write a SQL query to retrieve the first and last name of employees who work in the finance or marketing department, earn more than 52 000 and order the results by salary in descending order

```
Select   First_name,
         last_name,
         department,
         Salary
From     employee_db . employee
WHERE    (department = 'Finance' OR department = marketing)
         AND salary > 52 000 ORDER BY salary DESC;
```

Employee

first_name	last_name	department	Salary
Mike	Johnson	Finance	60 000
Robert	Wilson	Finance	59 000
Daniel	Clark	Marketing	53 000

12 Combining Select Distinct, WHERE, and IN

write a SQL query to find all the unique cities where employees work, excluding those in the IT and HR departments

```
Select Distinct cities
FROM employee_db.employee
WHERE department IN('Finance','Marketing');
```

City	department
Los Angeles	Finance
San Francisco	Marketing
Houston	Finance
Chicago	Marketing

13 Combining WHERE, NOT, AND and ORDERBY

Write a SQL enquery to retrieve employees who are NOT in finance department, have a salary greater than 50 000 and order the results by hire date in ascending order

```
Select  hire_date,
        department,
        Salary
FROM    employee_db.employee
WHERE   NOT department = finance
        AND salary > 50 000
ORDER BY hire_date ASC;
```

hire_date	department	Salary
2015-02-14	IT	62 000
2016-04-10	Marketing	52 000
2018-05-22	HR	51 000
2018-06-15	IT	55 000
2021-03-25	IT	53 000
2022-06-01	Marketing	53 000

14 Combining WHERE, OR, IN and LIMIT

Write a SQL query to find the first 3 employees who work in either Chicago or Los Angeles and belong to the IT or Marketing department

```
Select  first_name,
~~Select~~  City,
        Department
FROM    employee_db.employee
WHERE   (department = 'IT' OR department = 'Marketing') AND
        (city = 'chicago' OR city 'Los Angeles'),
LIMIT 3;
```

City	Department	first_name
Chicago	IT	Emily
Chicago	Marketing	Daniel